

Low-Hysteresis, High-Resolution Nonparallel Plate Capacitive Sensors for Soft Robot Shape Perception

Wenyan Zhang , Qing Cao , Pengfei Han , Guangda Qiao , Guofang Gong , Huayong Yang ,
Chun-Yi Su , *Senior Member, IEEE*, Fuzhou Niu , *Member, IEEE*,
Yueyue Liu , *Member, IEEE*, and Dong Han 

Abstract—The primary task of precise robot control is achieving shape perception. However, the nonlinear deformation and infinite degrees of freedom characteristics of soft robots make shape perception particularly challenging. Existing shape perception methods often suffer from strong dependencies on environmental conditions and motion constraints. This article reports a shape perception solution based on capacitance sensors, enabling three-dimensional shape reconstruction of a soft bellow robot. The proposed capacitive sensors adopt a nonparallel plate configuration with air as the dielectric and a wedge-shaped air gap between electrodes, whose geometry varies with robot deformation. The sensors are fabricated through a simple casting process, in which cured silver nanowire-based electrodes are embedded into polydimethylsiloxane substrates. For shape perception, multiple capacitive sensors are circumferentially distributed on the bellow surface to capture deformation-induced local inclination and reconstruct the global three-dimensional shape. Experimental results demonstrate that the nonparallel plate capacitance sensor exhibits low hysteresis ($\gamma_H \approx 1.70\%$), high

resolution (0.25°), and a fast response (218 ms). These attributes enable the sensor to perform precise and rapid shape perception under strict experimental environments. Furthermore, the combination of distributed multi-location capacitance measurements and a deep learning network successfully captures the posture information of the bellow, resulting in millimeter-level global shape reconstruction (with an average error of 2.547 ± 1.230 mm on a 124 mm bellow robot). This drive-perception integrated system provides accurate shape information for soft robots operating in highly constrained environments, such as those encountered in endoscopic and minimally invasive medical procedures.

Index Terms—Capacitive sensors, sensor systems and applications, shape perception, soft robot.

I. INTRODUCTION

THE inherent rigidity of traditional rigid robots' structures leads to limitations in medical applications. These limitations are primarily manifested in their inability to adapt to complex, dynamic environments, and their failure to meet stringent safety standards [1], [2]. In contrast, composed of elastic components with infinite degrees of freedom, soft robots naturally possess safety and adaptability advantages in confined and complex environments [3], [4]. Particularly, controllable soft bellows robots with high flexibility demonstrate significant potential for applications in medical rehabilitation [5], [6], [7]. To enhance the precision of their manipulation, shape perception is critical for the control of soft bellows robots [8], [9], [10]. It aims to detect real-time signals, map them to posture information, and reconstruct three-dimensional shapes. However, the continuous material and nonlinear deformation characteristics of bellows actuators increase the difficulty of shape perception under high degrees of freedom [11], [12]. Furthermore, due to the inherent rigidity, traditional sensing methods, such as encoders [13], [14] and inertial measurement units (IMUs) [15], [16], limit the deformation of soft robots and make them unsuitable for direct application [9]. Therefore, developing deformable and flexible sensors for shape perception of soft devices, such as bellows, is of significant importance [17], [18].

Received 26 February 2026; revised 26 April 2026; accepted 17 May 2026. This research was supported in part by the National Key Research and Development Program of China under Grant 2022YFC3802302 and in part by the National Natural Science Foundation of China under Grant 52475075. (Corresponding authors: Fuzhou Niu; Dong Han.)

Wenyan Zhang, Qing Cao, Guangda Qiao, Guofang Gong, Huayong Yang, and Dong Han are with the School of Mechanical Engineering, Zhejiang University, Hangzhou 310058, China, and also with the State Key Laboratory of Fluid Power and Mechatronic Systems, Hangzhou 310058, China (e-mail: wenyan_zhang@zju.edu.cn; qing_cao@zju.edu.cn; guangdaqiao@zju.edu.cn; gfgong@zju.edu.cn; yhy@zju.edu.cn; dong_han@zju.edu.cn).

Pengfei Han is with the Faculty of Engineering Technology, University of Twente, 7522NB Enschede, The Netherlands (e-mail: p.han@utwente.nl).

Chun-Yi Su is with Gina Cody School of Engineering and Computer Science, Concordia University, Montreal, QC H3G 1M8, Canada (e-mail: chun-yi.su@concordia.ca).

Fuzhou Niu is with the School of Mechanical Engineering, Suzhou University of Science and Technology, Suzhou 215009, China, and also with the State Key Laboratory of Fluid Power and Mechatronic Systems, Zhejiang University, Hangzhou 310058, China (e-mail: fznui@usts.edu.cn).

Yueyue Liu is with the School of Internet of Things Engineering, Jiangnan University, Wuxi 214122, China (e-mail: yueyueliu@jiangnan.edu.cn).

Digital Object Identifier 10.1109/TIE.2026.3697445

To achieve shape perception in soft devices, current mainstream research approaches are broadly categorized into three types: machine vision, optical signal perception, and piezoresistive sensing. First, machine vision systems rely on image recognition for perception. Some studies [19], [20] integrated cameras into the robot's interior and used image information to enable shape perception. However, the limited field of view of the internally embedded camera hinders continuous tracking of the soft robot's tip during large deformations, leading to visual feature loss and the inability to perceive global positional information [21], [22]. Therefore, some researchers [23], [24] have employed external depth cameras or stereo camera systems to capture the complete 3-D point cloud of the robot. However, this method is also limited in scenarios with visual occlusions.

Second, the optical signal perception approach obtains shape information by using physical properties of light such as intensity, wavelength, and phase [25]. Regarding light intensity, Zhao et al. [26] developed a stretchable optical waveguide strain sensor based on the intensity loss characteristics during light transmission and integrated it into a soft prosthetic hand. In terms of light wavelength, fiber Bragg grating (FBG) is a waveguide structure that responds to specific wavelengths. Dong et al. [27] utilized multi-core FBG fibers for distributed sensing and control of soft robots, achieving a maximum tip error of 3.11 mm when reconstructing a single bending segment. For the light phase, Li et al. [28] reported a shape-sensing technology based on optical frequency domain reflectometry (OFDR) phase demodulation and enhanced the spatial resolution of perception. However, these methods are not only costly but also heavily dependent on the stability of certain optical characteristics, which impose strict requirements on the light source, external equipment, and the environment [25], [29], [30], [31].

Third, the piezoresistive sensing method operates based on the change in resistance of piezoresistive materials due to shape deformation [32]. Yan et al. [33] mixed silicone rubber with carbon fibers at different volume fractions and cured the mixture to fabricate soft actuators with sensing capabilities. Additionally, some studies [34], [35] have integrated piezoresistive sensors with actuation units to enhance the resolution under large bending. However, the elastic properties of piezoresistive materials lead to significant hysteresis and slow response times, resulting in larger errors at higher movement speeds [30].

The aforementioned methods have yielded extensive research outcomes. However, capacitive sensors with high accuracy, low hysteresis, and low-power consumption [36], [37], [38], [39] have rarely been applied to shape perception. Capacitive sensors detect changes in structural parameters, such as electrode size and gap, based on variations in capacitance. Although the existing literature on this topic is limited, Hu et al. [40] provided one of the rare contributions in this area. Hu et al. [40] employed an array of stretchable parallel-plate capacitive sensors to achieve shape perception in soft robots. However, due to the temperature dependence of the dielectric constant of the material between the plates, temperature will easily cause variations in the capacitance of the parallel plates. This leads to the capacitance perception scheme being particularly sensitive.

In this work, we propose a shape perception solution based on nonparallel plate capacitive sensors. The sensors are attached

to the V-shaped sidewall of the bellows robot. Therefore, the capacitive sensor perceives shape changes of the bellows by detecting capacitance variations due to sidewall angle changes during deformation. Moreover, the nonparallel plate structure causes air to be the dielectric material between the plates, which significantly alleviates the sensor temperature sensitivity arising from the temperature dependence of the dielectric medium. Since the sensors are integrated with the actuator, global self-perception is achieved in dark, enclosed environments. Compared to the complex manufacturing techniques of FBG sensors, the capacitive sensor also shows low hysteresis ($\gamma_H \approx 1.70\%$) and high resolution (0.25°). This approach significantly reduces system complexity and simplifies experimental conditions, without compromising sensing performance. Table I demonstrates the comparison of performance between this article and previous studies. In addition, the sensor exhibits a rapid response time (218 ms), which is close to the human reaction time limit. Subsequently, we propose a deep learning-based shape perception algorithm utilizing distributed nonparallel plate capacitance. The scheme achieved millimeter-level global 3-D shape reconstruction under complex motion with an average error of 2.547 ± 1.230 mm. Table II provides a comparison of the proposed perception system against other existing perception systems. With low hysteresis, high resolution, and great stability, this approach suits soft bellow robots' shape perception in dark or enclosed environments.

The rest of this article is organized as follows. Section II introduces the theoretical model of the capacitive shape perception scheme and the fabrication of capacitive sensors. Section III demonstrates the experiment of the sensor's performance and results. Section IV introduces the application of the capacitive shape perception scheme on soft bellows robots and compares it with a visual scheme. Finally, Section V presents the conclusions.

II. PERCEPTION METHOD AND SENSOR FABRICATION

A. Theoretical Model

The bellows actuator's shape perception and detection are crucial for the precise and closed-loop control of the actuator. In this study, we attached the capacitive angle sensors on a bellows actuator as shown in Fig. 1(a), which can detect the morphological change differentially. To simplify the mathematical model, this section considers the locomotion only on a plane. The sensors are fixed on both sides of the actuator oppositely so that they can respond to the movement in the sensors' plane. When the actuator bends and stretches, the outside capacitance will increase and the inside capacitance will decrease. The detailed formulation is introduced in Section II-B. To rebuild the shape of the actuator, we used the piecewise constant curve (PCC) model [27]. The model considered every bending section as a constant curve arc. As a result, the whole continuum actuator was modeled by several different arcs. This approach is simple and efficient, so that real-time control is possible.

Based on the PCC model, the actuator curve was split into n piecewise circular arcs and only bent in the XZ plane [48]. Regarding the fixed end as the original coordinate system, the sensors' positions were the nodes between two adjacent arcs.

TABLE I
COMPARISON OF THE PERFORMANCE PARAMETERS

Reference	Sensor	Deformation Mode	Hysteresis	Resolution
[32]	Electrolycra	Stretching	16%	/
[34]	Conductive polymer	Bending	/	<2.5°
[35]	Carbon fiber-based piezoresistive sensor	Bending	10%	<2.5°
[41]	Distributed fiberoptic sensor	Bending	/	0.5°
[42]	Thermoplastic elastomer optical fibers	Stretching	5%	1% strain
[43]	Hydrogel optical fibers	Stretching	/	1% strain
[38]	Capacitive strain sensor	Stretching	3%	/
This article	Nonparallel plate capacitance	Bending	1.70%	0.25°

TABLE II
COMPARISON OF THE PERCEPTION SYSTEMS

Reference	Actuator	Sensor Type	Algorithm	Output	Robot Length	Estimate Error
[44]	Pneumatic soft robot	Embedded camera	CNN	3-D point cloud	100 mm	MAE: 10.12 mm
[19]	Bellow actuator	Embedded camera	SVR	3-D tip position	140 mm	RMSE: 2.56–4.52 mm
[45]	Silicone plate	FBG sensor	ANN	Surface shape	45 mm	RMSE: 1.17 mm
[46]	2-DOF catheter	Multicore FBG sensor	ANN	3-D curves	40 mm	RMSE: 0.491 mm
[30]	Bellow actuator	Piezoresistive sensor	LSTM	3-D tip position	~70 mm	RMSE: 3.70–7.37 mm
[47]	Tentacle soft robot	Piezoresistive sensor	EKF	Curvature angle	/	RMSE: ~1°
This article	Bellow actuator	Capacitance sensor	3-D CNN	3-D curves	124 mm	RMSE: 2.547 mm

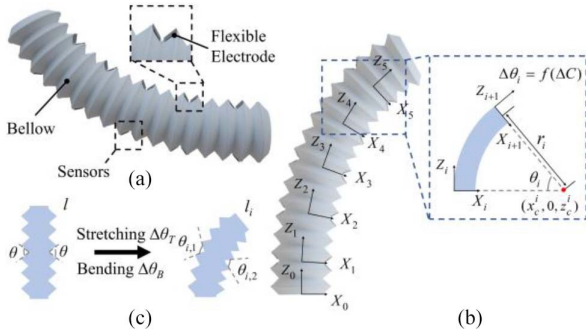


Fig. 1. Diagram and theoretical model of the capacitive method. (a) Schematic of the perception method. (b) PCC model. Configuration is divided into several segments. ΔC is the change in the sensor's capacitance values. (c) Diagram of the i th segment deformation with v-shaped side walls.

Consequently, the whole curve was split by $n - 1$ sensors. Meanwhile, both ends of the arcs were associated with a coordinate system. The radius and central angle of arcs are r_i and θ_i , and the center coordinates of arcs are $(x_c^i, 0, z_c^i)$ in the i th coordinate system, $i = 0, 2, \dots, n - 1$. The mathematical model is shown in Fig. 1(b). According to the parameters shown in the figure, the transformation matrix from the i th coordinate system to the $i + 1$ th system is determined by

$$T_i^{i+1} = \begin{bmatrix} R_i^{i+1} & d_i^{i+1} \\ 0^T & 1 \end{bmatrix} \quad (1)$$

where the rotation matrix R_i^{i+1} and translation matrix d_i^{i+1} are

$$R_i^{i+1} = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \end{bmatrix} \quad (2)$$

$$d_i^{i+1} = r_i \cdot \begin{bmatrix} 1 - \cos(\theta_i) \\ 0 \\ \sin(\theta_i) \end{bmatrix}. \quad (3)$$

Therefore, the transformation matrix from the original coordinate system to the i th system is determined by

$$T_i = \begin{bmatrix} R_i & d_i \\ 0^T & 1 \end{bmatrix}. \quad (4)$$

The rotation matrix R_i and translation matrix d_i are calculated iteratively by

$$R_i = R_{i-1} \cdot R_i^{i-1} \quad (5)$$

$$d_i = d_{i-1} + R_{i-1} \cdot d_i^{i-1} \quad (6)$$

where $R_0 = I$ and $d_0 = \mathbf{0}$. The position of any point in the $i + 1$ th arc is represented in the i th coordinate system as

$$\begin{cases} x' = x_c^i - r_i \cdot \cos(k \cdot \theta_i) \\ y' = 0 \\ z' = z_c^i + r_i \cdot \sin(k \cdot \theta_i) \end{cases}, k \in [0, 1] \quad (7)$$

where k denotes different points in the arc. The point could be represented in the original coordinate system by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = T_i \cdot \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}. \quad (8)$$

All points' curve positions could be obtained by θ_i , r_i , x_c^i , and z_c^i . However, the parameters can't be measured directly. We developed a learning method based on the sensors' capacitances. When the actuator bends or stretches, the angles of the bellows' V-shaped side wall are different. As a result, the difference in capacitances is generated. Therefore, using a data-driven approach, the central angle of arc θ_i is obtained by capacitances inversely. Then the radius r_i is calculated by

$$r_i = \frac{l_i}{\theta_i} \quad (9)$$

where l_i is the length of the arc, which is influenced by the actuator's extension or contraction. However, its locomotion is

coupled with bending and stretching. The angles of both V-shaped side walls can decouple their deformation. As shown in Fig. 1(c), the v-shaped angle is marked as $\theta_{i,1}$ and $\theta_{i,2}$ after stretching and bending. We assumed that the deformation is decomposed into stretching and then bending. First, the actuator extending or contracting, $\theta_{i,1}$ and $\theta_{i,2}$ will increase or decrease the same angle $\Delta\theta_T$. Then, after bending, $\theta_{i,1}$ and $\theta_{i,2}$ will increase or decrease $\Delta\theta_B$ separately. In the second process, the length of the arc will not change. $\theta_{i,1}$ and $\theta_{i,2}$ are represented by

$$\begin{cases} \theta_{i,1} = \theta + \Delta\theta_T + \Delta\theta_B \\ \theta_{i,2} = \theta + \Delta\theta_T - \Delta\theta_B \end{cases} \quad (10)$$

where θ is the initial angle of the wall. According to the assumption above, the length of the arc is only related to $\Delta\theta_T$. Hence, there is a relationship as (11), where l is the initial length, and l_i is the changed length. Then, r_i is determined.

$$l_i = f(\theta_{i,1} + \theta_{i,2}, l). \quad (11)$$

As shown in Fig. 1(b), in the theoretical case, within the i th coordinate system, x_c^i is the same as r_i , and z_c^i is approximately 0. Therefore, we aim to find a mapping from $\theta_{i,1}$ and $\theta_{i,2}$ to θ_i , r_i , x_c^i , and z_c^i , meanwhile resolving the nonlinear characteristic by a learning-based method. Consequently, the bellows actuator's shape curve is established.

The formula mentioned above involves coordinate transformations and matrix operations, making the iterative process complex. To reduce computational complexity, this study directly maps a series of capacitive values to the r_i , x_c^i , z_c^i , R_i , and d_i . However, due to inevitable errors in the mapping process, discontinuities may arise between adjacent arcs, resulting in a loss of smoothness in the generated curve. To address this issue, smoothness and continuity constraints are imposed on all curves during the generation process, thereby maximizing the effectiveness of the bellows curve shape generation. The specific loss function used is as follows:

$$\begin{aligned} \text{loss} = & \sum_M \| \mathbf{para}_{\text{optimize}} - \mathbf{para}_{\text{initial}} \| \\ & + \sum_N \| \mathbf{tangent}_{\text{end},i} - \mathbf{tangent}_{\text{start},i+1} \| \\ & + \sum_N \| \mathbf{point}_{\text{end},i} - \mathbf{point}_{\text{start},i+1} \|. \end{aligned} \quad (12)$$

Here, **para** represents the mapped parameters θ_i , r_i , x_c^i , and z_c^i , and the transformation matrices, while **tangent** refers to the unit tangent vector at the initial or final point of each arc segment, and **point** denotes the coordinates of these points.

The above theoretical model is established under the assumption that relative angular variations between the electrodes can be effectively detected through capacitance changes. When the angle becomes large (e.g., around 120° in this research), the sensitivity of the capacitance to angular changes decreases, which may weaken the identifiability of shape parameters inferred from capacitance variations. The bellows structure is therefore selected as the validation platform, as it provides moderate local initial angles that allow the sensor to operate

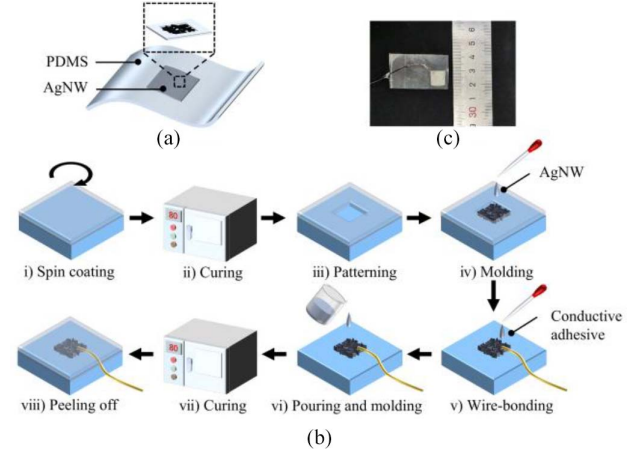


Fig. 2. Structure and fabrication of the flexible capacitive electrode. (a) Schematic of the capacitive electrode. (b) Schematic illustration of the fabrication process. (c) Photograph of the electrode.

within a high-sensitivity regime (e.g., 30° – 90° in this research) and facilitate verification of the proposed mapping model. The detailed quantitative sensitivity and performance results are provided in the Section III. For other segmented soft robots, similar performance can be achieved by appropriately selecting sensor attachment locations such that the sensor operates within an effective sensitivity range.

B. Sensor Fabrication

Nonparallel capacitive structures have been applied in studies on human motion sensing, force detection, and inertial measurement, demonstrating their capability to sense continuous deformation and micro geometric variations [49], [50], [51], [52], [53]. Based on these prior studies, this work adopts a flexible nonparallel plate capacitive structure for posture sensing of a soft bellows. Fig. 2(a) shows the schematic of the flexible capacitive electrode, in which the silver nanowire (AgNW) film was embedded in the polydimethylsiloxane (PDMS) membrane. The length and diameter of the nanowire are 70 nm and 35 μm , respectively. The fabrication process of the flexible electrode is shown in Fig. 2(b), inspired by previously reported fabrication methods for flexible electrodes [52], [54]. First, the PDMS was spin-coated onto a clean glass substrate at 1200 rpm for 30 s. Then, the PDMS membrane was cured at 80°C for 1 h. Subsequently, a $1 \times 1 \text{ cm}^2$ square hole was cut in the membrane and filled with the 3 wt% ethanol-based AgNW solution.

After the silver film was formed, a shielded cable was attached to the silver by a conductive adhesive to suppress parasitic capacitance from long wires. Following that, the PDMS was poured onto the silver and molded by a 1 mm deep mold. Finally, the electrode was cured at 80°C for 1 h and peeled off afterward. As shown in Fig. 2(c), the flexible capacitive electrode was trimmed to achieve the desired shape. The measured fabrication deviations of the electrode geometry were within sub-millimeter, while the PDMS thickness variation remained below 0.05 mm across all samples.

The flexible capacitive angle sensor consists of a pair of nonparallel electrodes. As long as one of the plates rotates, the capacitance of the sensor changes. Under idealized geometric assumptions, the capacitance–angle relationship of nonparallel electrodes can be formulated as [52], [53]

$$C = \frac{\varepsilon_0 \varepsilon_r b}{\theta} \ln \left(1 + \frac{l}{r} \right) \quad (13)$$

where ε_0 , ε_r , b , θ , l , and r are the permittivity of vacuum, the relative dielectric constant of the dielectric material, the width of the electrode, the angle between two inclined electrodes, the length of the electrode, and the radial distance from the rotation center to the edge of the electrode closest to the rotation axis, respectively. It should be noted that this formulation is derived based on simplified assumptions on geometry and electric field distribution. In practical implementations, factors, such as finite electrode size, variation of effective overlapping area, and edge-field effects, may cause deviations from this idealized model.

III. SENSOR PERFORMANCE

A. Experiment Setup

This section demonstrates the sensor's performance experiment and its results. The sensor's capacitance was measured by AD7746, which has two input channels with high resolution. The data acquisition circuit to obtain the sensor performance is shown in Fig. 3(a). The acquisition program was burned to an Arduino Uno to control the AD7746's work modes, which communicated with the AD7746 via inter-integrated circuit (I²C). The computer was used to monitor capacitance data and transmit commands to the Arduino. Fig. 3(b) presents the experimental setup. After building the circuit, the two electrodes of the sensor were separately fixed on the inside of hinges made by 3-D printing. To minimize assembly induced misalignment, the hinge structures were fabricated with auxiliary alignment features. The resulting positional offsets were within the sub-millimetre range. One of the hinges was fixed on the table, and another one was attached to the rotating test bench. The initial angular deviations were within $\pm 1.5^\circ$. A stepper motor drove the hinges to rotate so that capacitance values of various angles were measured. The motor has 1600 pulses per revolution, with a reduction ratio of 90:1. The pulse count for the rotation angle is calculated and then input into the controller to control the sensor's angle. In addition, to accurately measure the response and relaxation time of the sensor, the AD7746 was replaced by the FDC2214, which provides a maximum output rate of 4.08 kilo-samples per second (kSPS). The capacitance data were acquired by an STM32F103ZET6 at a sampling rate of 3.75 kHz.

B. Experiment Results

The sensor's response curve can be utilized to select an appropriate operating range and to determine the sensor's sensitivity. Fig. 4(a) shows the relative capacitance change $\Delta C/C_0$ in response to a sensor's angle, ranging from 30° to 120° , with a step size of 5° , and each step is tested 6 times. The capacitance and

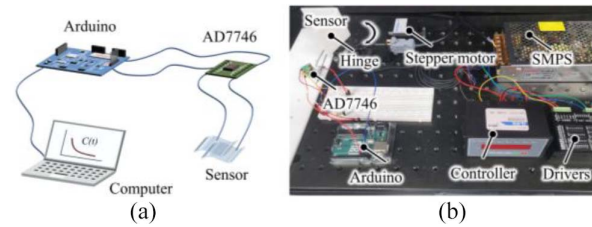


Fig. 3. Experiment platform. (a) Data acquisition circuit. (b) Experimental setup.

angle have an inverse relationship overall, which the response curve has validated. The sensitivity of the capacitance angle sensor is defined as $S = \Delta(C/C_0)/\Delta\alpha$, where C_0 is the reference capacitance, C denotes the measured capacitance, and $\Delta\alpha$ is the change of angle. The response curve should be divided into two parts because of inverse proportionality. The sensitivity S_1 is -0.0421 deg^{-1} for the range of 30° to 65° , and S_2 is -0.0066 deg^{-1} for the range of 75° to 120° approximately. Therefore, when perceiving larger angular changes, the sensor sensitivity will be reduced. The orange curve represents the transformed C_0/C data. It can be observed that the relationship between capacitance and angular variation approximately follows the first-order inverse proportional relationship described in (13). The hysteresis loop is shown in Fig. 4(b). The experimental parameters are set the same as in the previous case, except that the testing process involves increasing the angle first, followed by decreasing it in a single cycle. The response of the sensor, when the angle decreased and increased, is indicated by the green line and the orange line. According to the definition, hysteresis is expressed as $\gamma_H = \Delta H_{\max}/Y_{FS} \approx 1.70\%$. The sensor is not influenced by the shape memory of the material's elasticity during operation. Therefore, the hysteresis caused by different stroke directions has a minimal impact on the sensor's measurement performance.

As shown in Fig. 4(c), the sensor responses at several bending angles are measured over a certain period to evaluate measurement stability. For each angle, the capacitance signal is recorded for 10 s with a sampling rate of 1 Hz. The capacitance responses exhibit low variation at each fixed angle, indicating stable sensor output under static conditions. Sensors are prone to fatigue and degradation over extended periods of operation, which can adversely affect their performance. The ability to maintain reliable functionality over the long term is also a key evaluation criterion. Fig. 4(d) exhibits the sensor's robustness for 1000 cycles with an angle between 50° and 60° . Because the sensor does not exert tensile or bending stress on the plates during operation, which could lead to fatigue or damage, it is theoretically posited that the sensor possesses an extended operational lifespan. Furthermore, the curves representing the initial fifty cycles and the final fifty cycles are illustrated in the figure, demonstrating that the signal remains largely invariant. To assess the sensor stability over longer time scales, day-level aging tests were performed. As illustrated in Fig. 4(e), the relative capacitance of the sensor was recorded every two days over 10 days at three representative bending angles (120° , 60° , and 30°). The maximum deviation was approximately 0.008

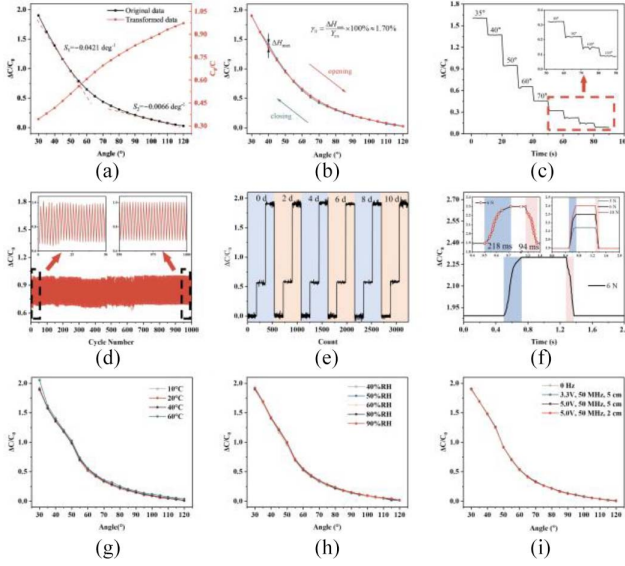


Fig. 4. Sensing performances of the sensors whose shape is a square with a side length of 10 mm. (a) Capacitance response test. (b) Hysteresis error of sensors. (c) Real-time capacitance response subjected to step angle change. (d) Stability of the sensor when the angle changed between 60° and 50° for 1000 cycles. (e) Capacitance responses at bending angles of 120°, 60°, and 30° over 10 days. (f) Dynamic response of the capacitive sensor at 30°. Upper-left: Response and relaxation time under a 6 N load. Upper-right: Sensor response under different applied loads (3, 6, and 10 N). (g) Capacitive sensor response curves at different temperatures. (h) Capacitive sensor response curves at different humidity. (i) Capacitive sensor response curves under different EMI conditions and distances.

(~0.42%). These results demonstrate that the proposed sensor maintains good stability under multi-day operation. Apart from the performance mentioned above, response time is another critical factor for applications of the sensor. When the electrode substrates come into contact, local compression and deformation occur in the flexible structure, which leads to larger response and relaxation times at the minimum angle. Fig. 4(f) indicates a response time of 218 ms and a relaxation time of 94 ms, under a constant external force of 6 N at a bending angle of 30°. Moreover, the response and relaxation times remain nearly unchanged under different external forces (3, 6, and 10 N), as shown in the upper-right subfigure of Fig. 4(f). It is worth noting that the reported value corresponds to the maximum response time under contact conditions. In noncontact operation, where contact-induced mechanical deformation effects are absent, the actual response time of the sensor is significantly shorter than 218 ms. Finally, the robustness of the sensor was evaluated under different environmental conditions. Fig. 4(g) shows the sensor response curves at different temperatures. In the noncontact bending range from 35° to 120°, the responses at different temperatures exhibit a small and nearly uniform overall offset (~0.046 at 60 °C), which can be compensated through zero-point recalibration or temperature compensation. At 60 °C and a bending angle of 30°, a significantly larger deviation in capacitance is observed (~0.163). This behavior is attributed to the contact between the sensor substrates, which induces local deformation and significantly increases the temperature sensitivity of the capacitive response. Therefore,

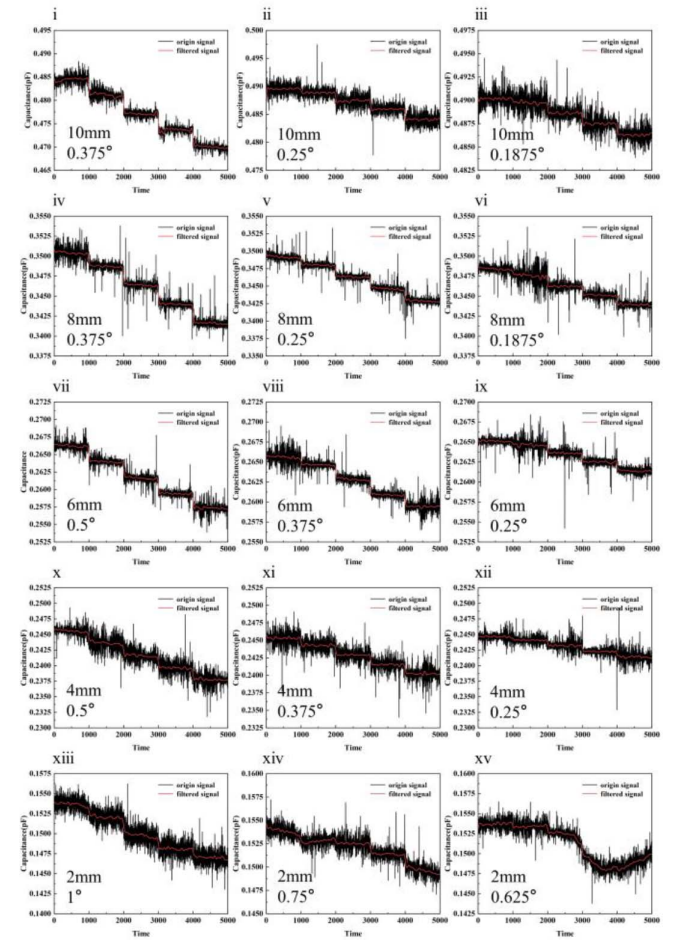


Fig. 5. Resolution testing of the sensors with different side lengths around 60°. Black: original signal; red: Signal after noise reduction using a butterworth low-pass filter. Two values shown in the lower left corner of each subplot indicate the electrode plate side length (top) and the stepwise angular increment applied after each set of 1000 recorded data points (bottom).

in practical applications, the electrode layout can be optimized to avoid substrate contact within the operating range, thereby reducing the influence of temperature on the sensor output. Fig. 4(h) and (i) further demonstrates that the sensor maintains excellent stability under different humidity conditions as well as under artificial electromagnetic interference (EMI) sources, where square-wave EMI signals with varying amplitudes and distances were applied.

The capacitance of the sensor is influenced by the shape and dimensions of the electrodes. When the same angular change occurs, sensors of varying sizes will yield different output values. However, due to the constraints imposed by the measurement system's resolution, the minimum angular resolution differs among sensors of various sizes. To assess and reinforce its applicability in practical scenarios, the minimum angular resolution of sensors with different side lengths was measured at an angle of 60° and subsequently compared, as illustrated in Fig. 5. Each sensor is rotated through a specified angle five times, with 1000 data points recorded during each rotation. The obtained data are then subjected to filtering. The employed filter

is a fourth-order Butterworth filter with a normalized cutoff frequency of 0.02. Following the experimental configuration, the sampling frequency is set to 1 Hz. The original signal and filtered signal are indicated by the black line and red line, respectively.

The minimum resolution test of the capacitive sensor with a 10 mm plate side length is shown in Fig. 5(i)–(iii), where the angle changes are 0.375° , 0.25° , and 0.1875° , respectively. When the angle change is 0.375° or 0.25° , the sensor can easily detect variations. However, when the angle change is reduced to 0.1875° , the first variation cannot be distinguished. Therefore, the minimum resolution of the 10 mm capacitive sensor, which is approximately 0.25° , is superior to that of distributed optical fiber sensors in terms of bending resolution (0.5°) [41]. Similarly, Fig. 5(iv)–(vi) show the response of the 8 mm capacitive sensor to the same angle changes. Its performance does not show significant differences compared to the 10 mm sensor. However, when the plate side length is reduced to 6 mm and 4 mm, the sensor performance deteriorates, as shown in Fig. 5(vii)–(ix) and (x)–(xii). The test results indicate that both sensors can only provide effective feedback for angle changes of 0.375° or greater, with occasional difficulty in clearly distinguishing the boundary of changes when the angle variation is 0.25° . It is well known that the capacitance value is proportional to the area of the electrode plates. As the plate size decreases, the capacitance value also decreases, leading to smaller capacitance variations for the same angular change. When measured with chips having the same resolution, smaller capacitors typically require larger angular changes to detect capacitance variations. Fig. 5(xiii)–(xv) further corroborates this point, where the 2 mm side length sensor can only reliably detect angle changes of 1° , with a significant decline in performance compared to sensors with larger side lengths. When the angle change is 0.625° , the curve even shows considerable distortion due to the inherent fluctuations in the measurement. In conclusion, the resolution of the sensor is compromised when used in narrow or miniature scenarios. Generally, a capacitive sensor with a 4 mm side length strikes a balance between sensor performance and suitability for testing in narrow environments.

IV. APPLICATION OF THE SENSOR SYSTEM

A. Experiment Setup

The sensor's performance ensures the feasibility of the perception scheme. This section presents the specific application of the proposed perception scheme on the bellows actuator. Fig. 6(a) shows the sensor layout on the sidewall of the bellows actuator. Three pairs of electrodes are symmetrically and differentially attached at 120° intervals along the circumferential direction of the actuator to sense shape changes in different directions within the space. Additionally, based on the length of the bellow, the electrode array along the circumference extends axially to segmentally measure the deformation of the actuator. To more realistically simulate real-world applications, two servos are used to pull the actuator at different heights and in different directions, creating more complex and

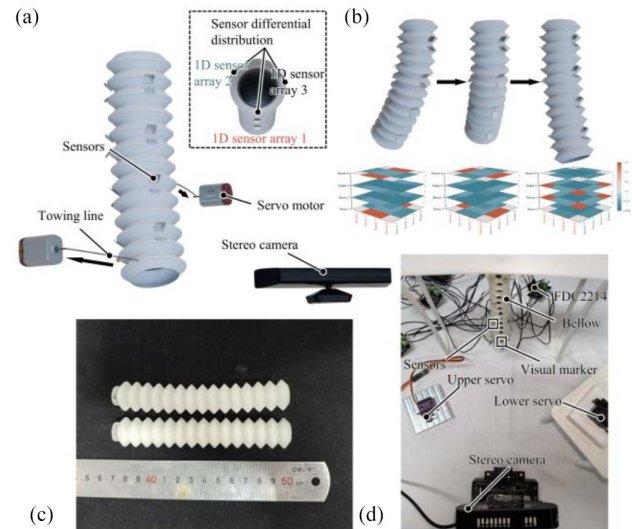


Fig. 6. Illustration of the shape perception principle and experimental setup for the bellows actuator. (a) Experimental setup for shape perception application, with the sensor distribution shown within the dashed box, consisting of four layers of sensor arrays. (b) Stacked diagram demonstrating the principle of shape perception, where different shapes of the bellows correspond to unique stacked diagrams in space. (c) Silicone-made bellows used in the application experiment. (d) Experimental setup.

multidimensional deformations in space. The experimental results are then compared with binocular vision measurements.

As shown in Fig. 6(b), when the bellow bends in different directions, the capacitance values of the different sensors vary. The stacked heat map in Fig. 6(b) is constructed by multiplying the capacitance values from different sensors. The different axes represent the different sensor arrays along the axial direction, and the axis colors correspond to the top view in Fig. 6(a). It can be observed that the different postures of the bellow correspond to different stacked heat maps. In other words, the bellows' posture changes are predicted and regressed through the capacitance stacking maps. The silicone bellow used in the experiment is shown in Fig. 6(c), with a length of 124 mm, a diameter ranging from 14 mm to 22 mm, a wall thickness of 1.5 mm, and 15 corrugation rings. The experimental setup is shown in Fig. 6(d), where the bellow is driven by two MG90S servos at the top and bottom through the towing line. A total of 12 capacitive sensors are integrated on each bellow, uniformly distributed circumferentially at 120° intervals and axially arranged in four evenly spaced layers. These sensors are used to capture the bending state of the bellow, and the 12 sensors effectively divide the soft robot into three axial segments, enabling the reconstruction of its complex three-dimensional shape. The capacitive sensors consist of 4 mm electrodes, as determined in Section III of this article, and the capacitance values are measured using the FDC2214. Binocular vision is implemented through the Orbbec Gemini Pro, with black markers attached to each corrugation of the bellow to reconstruct its spatial configuration.

B. Binocular Vision Measurement

In this section, a binocular vision approach is used to capture the bellow's shape under different loading conditions.

The resulting shape data are obtained as reference information for training the proposed capacitive-based learning model. These data are used only for offline calibration and are not involved in the subsequent capacitive-based estimation and evaluation. The posture states of the bellow are configured by adjusting the rotation angles and the force application positions of MG90S servo motors, whose mechanical connections are described in Section IV-A. By varying the servo rotation angles, the effective magnitude of the applied load can be adjusted. Changing the attachment positions of the servos leads to different force distributions along the bellow, thereby producing a diverse set of bending configurations. As shown in Fig. 7(a), a total of nine posture states of the bellow are considered to cover typical deformation modes. State 1 corresponds to the undeformed configuration. States 2–5 represent postures generated by a single external load. By varying the load magnitude and the application position, different degrees of planar bending are produced. States 6–9 correspond to postures generated by the combined action of two external loads, where different combinations of load application positions and directions induce more complex asymmetric or nonplanar bending configurations. The force distribution and directions are indicated in the figure. These nine posture states are not intended to exhaust all possible deformations. Instead, they are used as an offline calibration set for learning the mapping between sensor signals and spatial shape. For each posture state, the spatial configuration of the bellow is measured using a binocular vision system. Specifically, the two-dimensional pixel coordinates of the visual markers are extracted from red-green-blue (RGB) images, while the corresponding depth information is obtained from depth images. By fusing the pixel coordinates with the depth data, the three-dimensional spatial position of each marker is reconstructed, yielding a set of discrete 3-D points distributed along the axial direction of the bellow. These points provide the input for subsequent shape modelling based on the PCC assumption.

Based on the PCC assumption described in Section II-A, the bellow centerline can be modeled using multiple circular arc segments, where each arc segment is approximated as a constant-curvature bend contained within a local plane. The extracted 15 visual points are ordered along the axial direction and divided into three groups. Each group is then used to fit an arc segment for modelling the axial posture of the bellows. Fig. 7(b) illustrates the spatial distribution and grouping of the visual points, as well as the corresponding arc planes determined by each group. Points with different colors represent different groups, and the fitting plane corresponding to each group is indicated using the same color. It should be noted that Fig. 7(b) presents only State 3 as a representative example to illustrate the grouping and plane-fitting procedure. The same processing pipeline is applied to all other posture states. Each group of points is used to determine an initial arc using the least squares method. To ensure geometric consistency, continuity and smoothness constraints between the arcs are applied to iteratively generate the final fitting curve, as shown in Fig. 7(c). The black points denote the marker coordinates extracted by the binocular vision system, while the red curve represents the

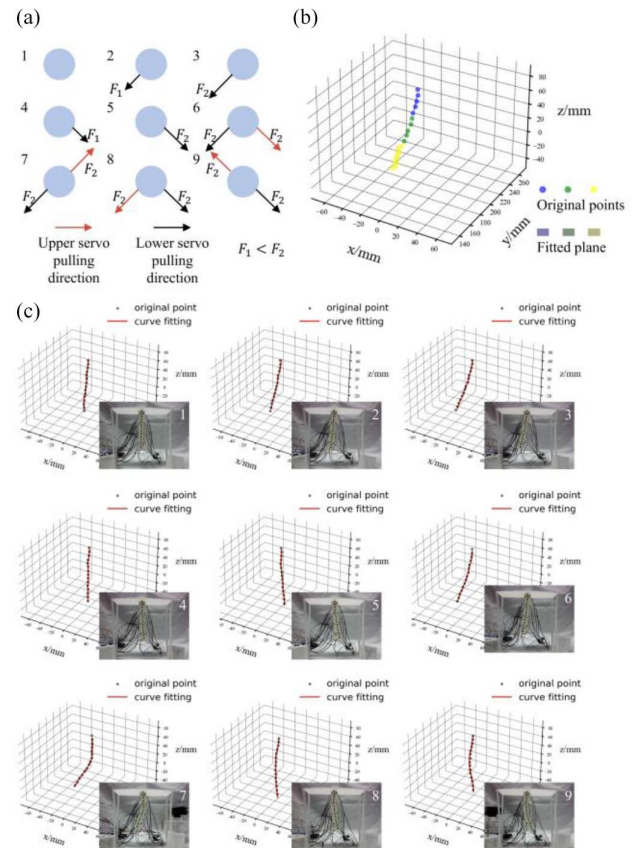


Fig. 7. Nine states of the bellows and visual perception scheme. (a) Nine states of the bellows and their corresponding force conditions. (b) Visual perception scheme extracts 15 key points, grouped into three sets of five, which are fitted into three arc segments. Different colors in the figure represent the points from different groups and the corresponding fitted arc planes. (c) Bellows morphology for the nine states fitted by the visual perception scheme, with the corresponding RGB image shown in the lower right corner.

final fitted centerline. To demonstrate the effectiveness of the fitting results, each subfigure corresponds to a specific bellow state, with the associated RGB image shown in the lower-right corner. The geometric parameters of the fitted arc segments, such as the center coordinates and radii, are subsequently used as supervision labels for training the learning model.

C. Capacitive Sensing Measurement

Subsequently, capacitive sensors are employed to sense and predict the posture of the bellows. For clarity, the capacitive array configuration, consisting of four circumferential layers of differential electrodes arranged at 120° intervals, is briefly summarized in Fig. 6(a). As a result, the bellow is equipped with 12 pairs of electrodes. During the data collection process, 1000 datasets are measured for each posture, leading to a total of 9000 datasets across the nine states. Each dataset includes 12 distinct capacitance values, and the resulting variation curves are shown in Fig. 8(a). The data sequence corresponds directly with the posture sequence shown in Fig. 7(a).

According to the content in Section II-A, the capacitance values are mapped to the bellow's posture. However, due to

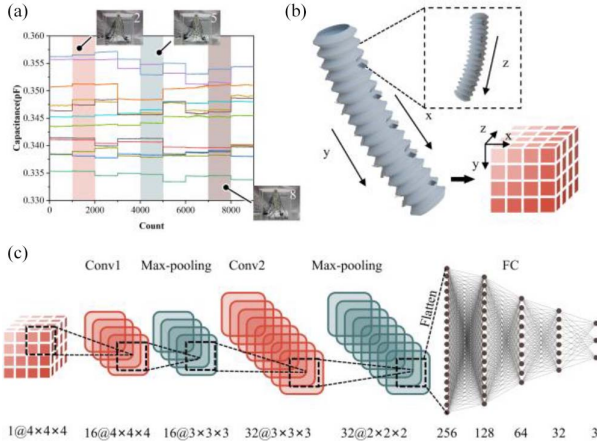


Fig. 8. Capacitive sensor-based shape perception scheme. (a) Measured capacitive data variation curves for 12 pairs of sensors under nine different morphologies. (b) Capacitive data preprocessing method. Three dimensions of capacitive data are converted into a three-dimensional data matrix, with matrix elements formed by multiplying the corresponding data from the three dimensions. (c) Three-dimensional convolutional neural network for perception. Output is the bellows morphology parameter.

the difficulty of constructing a nonlinear mathematical model for precise prediction, deep learning is employed for regression calculation. First, the capacitance array is transformed into the input for the training network. Similar to the previous processing approach, the capacitance values on the same side are normalized and converted into values of the same dimension. For example, $x_1, \dots, x_4, y_1, \dots, y_4, z_1, \dots, z_4$, are transformed into a three-dimensional matrix *Input*, where $Input(i, j, k) = x_i \times y_j \times z_k$, as shown in Fig. 8(b). The convolutional neural network structure is shown in Fig. 8(c), consisting of two convolutional layers and two max-pooling layers. The kernel size of the convolutional layer is $3 \times 3 \times 3$, the kernel size of the pooling layer is $2 \times 2 \times 2$, the stride for convolution is 1, and zero padding is applied. The processed results are then flattened and input into a fully connected layer with three hidden layers. The predicted values in the training dataset correspond to the features of three arcs fitted by the binocular visual scheme, including the x and y coordinates of the center, radius, the start angles, and the end angles of each arc segment.

The training loss is defined as the mean squared error (MSE) loss, with the stochastic gradient descent optimizer using momentum. For example, using the center's x-coordinates of the three arc segments, the hyperparameters of the network were optimized, and the results are shown in Table III. It can be seen that the lowest loss occurs when the learning rate is 0.02, the momentum is 0.08, the pooling layer is max pooling, and the activation function is rectified linear unit.

Using the same method as above, the center coordinates, radius, start and end angles of the arc, translation vector of the arc plane, and rotation matrix were trained. The rotation matrix is determined based on Rodrigues' rotation formula using the unit rotation axis vector and rotation angle, as (14) shows. $\mathbf{v}$ and $\mathbf{v}'$ are the original vector and the rotated vector, respectively. $\mathbf{u}$ is the unit rotation axis vector, while θ is the rotation angle

$$\mathbf{v}' = \mathbf{v} \cos(\theta) + (\mathbf{u} \times \mathbf{v}) \sin \theta + \mathbf{u}(\mathbf{u} \cdot \mathbf{v})(1 - \cos \theta). \quad (14)$$

TABLE III
HYPERPARAMETER OPTIMIZATION

Learning Rate	Momentum	Pooling Layer	Activation Function	Loss
0.01	0.08	Max pooling	ReLU	1.42×10^{-4}
0.02	0.08	Max pooling	ReLU	3.68×10^{-5}
0.03	0.08	Max pooling	ReLU	0.816
0.02	0.07	Max pooling	ReLU	1.08×10^{-4}
0.02	0.09	Max pooling	ReLU	5.25×10^{-5}
0.02	0.1	Max pooling	ReLU	0.829
0.02	0.08	Average pooling	ReLU	0.818
0.02	0.08	Max pooling	Softmax	0.818
0.02	0.08	Max pooling	Tanh	4.40×10^{-5}

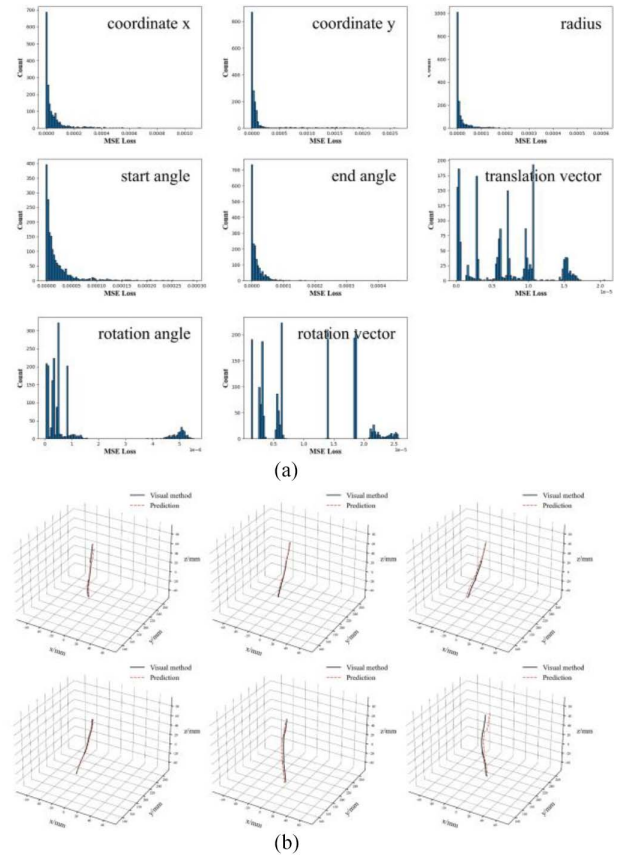


Fig. 9. Capacitive sensor-based shape perception results. (a) Bar chart of MSE loss for the morphology parameters. (b) Comparison between the capacitive sensor perception scheme and the visual perception scheme for six selected states. Black line represents the result of the visual perception scheme, and the orange dashed line represents the result of the capacitive perception scheme.

The bar chart of the MSE loss for all training results is shown in Fig. 9(a). The errors for coordinates, radius, and start/end angles are concentrated on the left side of the axis. The prediction results for the translation vector and rotation matrix, with a maximum error of 2.5×10^{-5} , are comparatively scattered, but the regression performance is still satisfactory.

To verify the fitting accuracy, six sets of data were measured. Based on the predicted results, arcs were drawn, and constraints were applied to ensure the smoothness and continuity between each arc segment. The final fitting curve was generated through

TABLE IV
PREDICTION ERROR

Status and Arcs	Prediction Error (mm)
First arc	2.98
1 Second arc	3.46
Third arc	3.94
First arc	0.92
2 Second arc	1.24
Third arc	1.92
First arc	1.04
3 Second arc	2.40
Third arc	2.92
First arc	1.17
4 Second arc	1.54
Third arc	2.54
First arc	3.08
5 Second arc	2.53
Third arc	2.04
First arc	5.91
6 Second arc	2.79
Third arc	3.42

iteration. As shown in Fig. 9(b), the black solid line represents the visual fitting result, and the orange dashed line represents the result predicted by the deep learning network regression. The error results are shown in Table IV, with an average error of 2.547 ± 1.230 mm and a maximum error of 5.91 mm.

Meanwhile, the computational cost and real-time capability of shape prediction are critical for practical applications. The proposed network adopts a lightweight design while maintaining high prediction accuracy, with a model size of 1.84 MB and approximately 3.55 million multiply-accumulate operations. During model evaluation, the memory footprint is about 10 MB, and the GPU utilization remains around 10% (RTX 4070), with a single inference time of 1.7426 ms. Including data acquisition, preprocessing, and post-processing, the total system latency is approximately 1.931 ms. In comparison, the mechanical bandwidth of typical soft actuators is generally below 20 Hz [55], indicating that the proposed perception system satisfies the real-time requirements in practical applications.

V. CONCLUSION

In this study, a morphology sensing method based on flexible nonparallel plate capacitive sensors is proposed to address the challenge of reliable shape measurement of soft robots. By distributing multiple pairs of capacitive units along the V-shaped sidewalls of the bellows robot, the proposed approach directly senses local angular variations. On this basis, a deep learning-based model is developed to globally reconstruct the three-dimensional morphology of the bellows using multi-channel capacitive signals. To validate the applicability of the proposed sensing scheme in soft robotic systems, both the sensor characteristics and the performance of the shape perception approach were experimentally evaluated. Static characterization results indicate that the proposed sensor exhibits low hysteresis ($\gamma_H \approx 1.70\%$) and sufficient angular resolution (0.25°), enabling accurate angle measurement between 30° and 120° and providing reliable inputs for shape prediction. Dynamic response tests further demonstrate a fast response time (218 ms), which satisfies the requirements for real-time morphology feedback of soft bellows. In addition, repeated bending, lifetime, and

extreme-environment tests show that the sensor maintains stable output under cyclic deformation, long-term operation, and environmental disturbances. These validate its robustness and reliability for long-term deployment in soft robotic applications. To assess the accuracy of the capacitance-based sensing approach, an evaluation of the proposed perception model was conducted. The system provides stable reconstruction results across various deformation morphologies of the bellows. Compared with vision-based measurements, it achieves an average reconstruction error of 2.547 ± 1.230 mm for a 124-mm-long bellows, achieving accuracy similar to that of the FBG method (>2 mm [45]). The experimental results demonstrate that the approach enables effective morphology sensing for soft robots, offering advantages such as suitability for confined spaces, structural simplicity, and good compliance. Therefore, it has the potential to enhance minimally invasive surgery and endoscopic-assisted procedures in medical applications.

In the future, the flexible capacitive sensor will be embedded into soft robotic structures through in-mold curing and stretchable wiring techniques. This is expected to enhance system integration and enable a broader range of motion detection capabilities.

REFERENCES

- [1] J. Burgner-Kahrs, D. C. Rucker, and H. Choset, "Continuum robots for medical applications: A survey," *IEEE Trans. Robot.*, vol. 31, no. 6, pp. 1261–1280, Dec. 2015.
- [2] H. Cheng, L. Chen, B. Fang, J. Zhang, and J. Hong, "Switching control for a soft rehabilitation glove with pneumatic bellows actuators," *IEEE Robot. Autom. Lett.*, vol. 9, no. 6, pp. 4966–4973, Jun. 2024.
- [3] B. Yi, Y. Fan, D. Liu, and J. Guadalupe Romero, "Simultaneous position-and-stiffness control of underactuated antagonistic tendon-driven continuum robots," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 7238–7254, 2025.
- [4] Y. Fei and H. Xu, "Modeling and motion control of a soft robot," *IEEE Trans. Ind. Electron.*, vol. 64, no. 2, pp. 1737–1742, Feb. 2017.
- [5] Z. Chen et al., "Data-driven methods applied to soft robot modeling and control: A review," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 2241–2256, 2025.
- [6] J. Xia et al., "3D-printed passive bellow actuator for portable soft wearable robots," *Smart Mater. Struct.*, vol. 33, no. 4, Mar. 2024, Art. no. 045018.
- [7] Z. Jiryaee and A. S. Jafarpisheh, "The usefulness of assistive soft robotics in the rehabilitation of patients with hand impairment: A systematic review," *J. Bodyw. Mov. Ther.*, vol. 39, pp. 398–409, Jul. 2024.
- [8] X. An et al., "Shape reconstruction of soft continuum robots via the fusion of local strains and global poses," *Cell Rep. Phys. Sci.*, vol. 5, no. 10, Oct. 2024, Art. no. 102224.
- [9] M. Ilami, H. Bagheri, R. Ahmed, E. O. Skowronek, and H. Marvi, "Materials, actuators, and sensors for soft bioinspired robots," *Adv. Mater.*, vol. 33, no. 19, May 2021, Art. no. 2003139.
- [10] C. Shi et al., "Shape sensing techniques for continuum robots in minimally invasive surgery: A survey," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 8, pp. 1665–1678, Aug. 2017.
- [11] M. Sudani, M. Deng, and S. Wakimoto, "Modelling and operator-based nonlinear control for a miniature pneumatic bending rubber actuator considering bellows," *Actuators*, vol. 7, no. 2, Jun. 2018, Art. no. 2.
- [12] G. Goran et al., "Position control of a cost-effective bellow pneumatic actuator using an LQR approach," *Actuators*, vol. 12, no. 2, Feb. 2023, Art. no. 73.
- [13] R. J. Webster and B. A. Jones, "Design and kinematic modeling of constant curvature continuum robots: A review," *Int. J. Robot. Res.*, vol. 29, no. 13, pp. 1661–1683, Nov. 2010.
- [14] D. B. Camarillo, K. E. Loewke, C. R. Carlson, and J. K. Salisbury, "Vision based 3-D shape sensing of flexible manipulators," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, 2008, pp. 2940–2947.

- [15] H. Huang, C. Pang, F. Sun, M. Dong, Z. Wen, and H. Zhou, "Dexterity operation, multi-modal perception and tactile force interaction of a bionic soft hand," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 4, pp. 7219–7229, Oct. 2024.
- [16] W. Liu et al., "An inchworm-snake inspired flexible robotic manipulator with multisection SMA actuators for object grasping," *IEEE Trans. Ind. Electron.*, vol. 70, no. 12, pp. 12616–12625, Dec. 2023.
- [17] M. C. Yuen, H. Tonoyan, E. L. White, M. Telleria, and R. K. Kramer, "Fabric sensory sleeves for soft robot state estimation," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, 2017, pp. 5511–5518.
- [18] L. Wang et al., "Soft robot proprioception using unified soft body encoding and recurrent neural network," *Soft Robot.*, vol. 10, no. 4, pp. 825–837, Aug. 2023.
- [19] P. P. Werner, M. Hofer, C. Sferrazza, and R. D'Andrea, "Vision-based proprioceptive sensing: Tip position estimation for a soft inflatable bellow actuator," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2020, pp. 8889–8896.
- [20] R. Wang, S. Wang, S. Du, E. Xiao, W. Yuan, and C. Feng, "Real-time soft body 3D proprioception via deep vision-based sensing," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3382–3389, Apr. 2020.
- [21] R. Reilink, S. Stramigioli, and S. Misra, "3D position estimation of flexible instruments: Marker-less and marker-based methods," *Int. J. Comput. Assisted Radiol. Surgery*, vol. 8, pp. 407–417, May 2013.
- [22] P. Cabras, D. Goyard, F. Nageotte, P. Zanne, and C. Doignon, "Comparison of methods for estimating the position of actuated instruments in flexible endoscopic surgery," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2014, pp. 3522–3528.
- [23] J. Lai, B. Lu, and H. K. Chu, "Variable-stiffness control of a dual-segment soft robot using depth vision," *IEEE/ASME Trans. Mechatron.*, vol. 27, no. 2, pp. 1034–1045, Apr. 2022.
- [24] J. M. Croom, D. C. Rucker, J. M. Romano, and R. J. Webster, "Visual sensing of continuum robot shape using self-organizing maps," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, 2010, pp. 4591–4596.
- [25] X. Wang, Z. Li, and L. Su, "Soft optical waveguides for biomedical applications, wearable devices, and soft robotics: A review," *Adv. Intell. Syst.*, vol. 6, no. 1, 2024, Art. no. 2300482.
- [26] H. Zhao, K. O'Brien, S. Li, and R. F. Shepherd, "Optoelectronically innervated soft prosthetic hand via stretchable optical waveguides," *Sci. Robot.*, vol. 1, no. 1, Dec. 2016, Art. no. eaai7529.
- [27] Z. Dong et al., "Shape tracking and feedback control of cardiac catheter using MRI-guided robotic platform—Validation with pulmonary vein isolation simulator in MRI," *IEEE Trans. Robot.*, vol. 38, no. 5, pp. 2781–2798, Oct. 2022.
- [28] S. Li et al., "Twist compensated, high accuracy and dynamic fiber optic shape sensing based on phase demodulation in optical frequency domain reflectometry," *Mech. Syst. Signal Process.*, vol. 216, Jul. 2024, Art. no. 111462.
- [29] C. Wu, X. Liu, and Y. Ying, "Soft and stretchable optical waveguide: Light delivery and manipulation at complex biointerfaces creating unique windows for on-body sensing," *ACS Sens.*, vol. 6, no. 4, pp. 1446–1460, Apr. 2021.
- [30] J. Shu, J. Wang, K. C.-C. Cheng, L.-F. Yeung, Z. Li, and R. K. Tong, "An end-to-end dynamic posture perception method for soft actuators based on distributed thin flexible porous piezoresistive sensors," *Sensors*, vol. 23, no. 13, Jan. 2023.
- [31] U. Jeong and K.-J. Cho, "Control of a bowden-cable actuation system with embedded BoASensor for soft wearable robots," *IEEE Trans. Ind. Electron.*, vol. 67, no. 9, pp. 7669–7680, Sep. 2020.
- [32] M. Cianchetti, F. Renda, A. Licofonte, and C. Laschi, "Sensorization of continuum soft robots for reconstructing their spatial configuration," in *Proc. IEEE RAS & EMBS Int. Conf. Biomed. Robot. BioMechatronics (BioRob)*, 2012, pp. 634–639.
- [33] X. Yan, C. R. Bowen, C. Yuan, Z. Hao, and M. Pan, "Carbon fibre based flexible piezoresistive composites to empower inherent sensing capabilities for soft actuators," *Soft Matter*, vol. 15, no. 40, pp. 8001–8011, Oct. 2019.
- [34] J. Zhou, Y. Chen, X. Chen, Z. Wang, Y. Li, and Y. Liu, "A proprioceptive bellows (PB) actuator with position feedback and force estimation," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 1867–1874, Apr. 2020.
- [35] M. Pan et al., "Soft controllable carbon fibre-based piezoresistive self-sensing actuators," *Actuators*, vol. 9, no. 3, Aug. 2020, Art. no. 79.
- [36] W. Gao et al., "A high-resolution mems capacitive force sensor with bionic swallow comb arrays for ultralow multiphysics measurement," *IEEE Trans. Ind. Electron.*, vol. 70, no. 7, pp. 7467–7477, Jul. 2023.
- [37] U.-H. Shin, D.-W. Jeong, S.-M. Park, S.-H. Kim, H. W. Lee, and J.-M. Kim, "Highly stretchable conductors and piezocapacitive strain gauges based on simple contact-transfer patterning of carbon nanotube forests," *Carbon*, vol. 80, pp. 396–404, Dec. 2014.
- [38] D. J. Cohen, D. Mitra, K. Peterson, and M. M. Maharbiz, "A highly elastic, capacitive strain gauge based on percolating nanotube networks," *Nano Lett.*, vol. 12, no. 4, pp. 1821–1825, Apr. 2012.
- [39] S. Yao, J. Yang, F. R. Pobleto, X. Hu, and Y. Zhu, "Multifunctional electronic textiles using silver nanowire composites," *ACS Appl. Mater. Interfaces*, vol. 11, no. 34, pp. 31028–31037, Aug. 2019.
- [40] D. Hu, F. Giorgio-Serchi, S. Zhang, and Y. Yang, "Stretchable e-skin and transformer enable high-resolution morphological reconstruction for soft robots," *Nature Mach. Intell.*, vol. 5, no. 3, pp. 261–272, Mar. 2023.
- [41] H. Bai, S. Li, J. Barreiros, Y. Tu, C. R. Pollock, and R. F. Shepherd, "Stretchable distributed fiber-optic sensors," *Science*, vol. 370, no. 6518, pp. 848–852, Nov. 2020.
- [42] A. Leber et al., "Stretchable thermoplastic elastomer optical fibers for sensing of extreme deformations," *Adv. Funct. Mater.*, vol. 29, no. 5, Feb. 2019, Art. no. 1802629.
- [43] J. Guo et al., "Highly stretchable, strain sensing hydrogel optical fibers," *Adv. Mater.*, vol. 28, no. 46, pp. 10244–10249, 2016.
- [44] U. Yoo, H. Zhao, A. Altamirano, W. Yuan, and C. Feng, "Toward zero-shot sim-to-real transfer learning for pneumatic soft robot 3D proprioceptive sensing," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, London, U.K., 2023, pp. 544–551.
- [45] T. L. Lun, K. Wang, J. D. L. Ho, K.-H. Lee, K. Y. Sze, and K.-W. Kwok, "Real-time surface shape sensing for soft and flexible structures using fiber Bragg gratings," *IEEE Robot. Autom. Lett.*, vol. 4, no. 2, pp. 1454–1461, Apr. 2019.
- [46] X. T. Ha et al., "Shape sensing of flexible robots based on deep learning," *IEEE Trans. Robot.*, vol. 39, no. 2, pp. 1580–1593, Apr. 2023, doi: 10.1109/TRO.2022.3221368.
- [47] J. Y. Loo, K. C. Kong, C. P. Tan, and S. G. Nurzaman, "Non-linear system identification and state estimation in a pneumatic based soft continuum robot," in *Proc. IEEE Conf. Control Technol. IEEE Int. Symp. Spread Spectr. Tech. Appl. (CCTA)*, 2019, pp. 39–46.
- [48] F. Stella, Q. Guan, C. D. Santina, and J. Hughes, "Piecewise affine curvature model: A reduced-order model for soft robot-environment interaction beyond PCC," in *Proc. IEEE Int. Conf. Soft Robot. (RoboSoft)*, 2023, pp. 1–7.
- [49] L. Huang et al., "A flexible detachable capacitive sensor based on nonparallel electrodes for joints angle measurement," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–9, 2024.
- [50] S. Yu and H. Yu, "Capacitive stretchable strain sensor with low hysteresis based on wavy-shape interdigitated metal electrodes," *IEEE Sens. J.*, vol. 21, no. 24, pp. 27335–27342, Dec. 2021.
- [51] J.-F. Wu et al., "Human limb motion detection with novel flexible capacitive angle sensor based on conductive textile," *Electronics*, vol. 7, no. 9, Sep. 2018, Art. no. 9.
- [52] W. Song, T. Wang, and L.-H. Shao, "Analytical model for large deformation and capacitance of rhomboidal pressure sensors and experimental verification," *Acta Mech. Sin.*, vol. 39, no. 9, Jun. 2023, Art. no. 122497.
- [53] F. E. H. Tay, X. Jun, Y. C. Liang, V. J. Logeeswaran, and Y. Yufeng, "The effects of non-parallel plates in a differential capacitive microaccelerometer," *J. Micromechanics Microengineering*, vol. 9, no. 4, pp. 283–293, Dec. 1999.
- [54] H.-S. Liu, B.-C. Pan, and G.-S. Liou, "Highly transparent AgNW/PDMS stretchable electrodes for elastomeric electrochromic devices," *Nanoscale*, vol. 9, no. 7, pp. 2633–2639, Feb. 2017.
- [55] B. Gorissen, D. Reynaerts, S. Konishi, K. Yoshida, J. Kim, and M. De Volder, "Elastic inflatable actuators for soft robotic applications," *Adv. Mater.*, vol. 29, no. 43, Nov. 2017, Art. no. 1604977.



Wenyan Zhang received the B.E. degree in mechanical engineering from the College of Mechanical and Electrical Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China. He is currently working toward the Ph.D. degree in mechatronic engineering with the School of Mechanical Engineering, Zhejiang University, Zhejiang, China.

His research interests include soft robotics and medical equipment.



Qing Cao received the B.E. degree in mechanical engineering from the School of Mechanical Engineering, Wuhan University of Technology, Wuhan, China, in 2020. He is currently working toward the Ph.D. degree in mechatronic engineering with the School of Mechanical Engineering, Zhejiang University, Zhejiang, China.

His research interest includes the application of magnetic drive technology.



Pengfei Han received the B.Eng. and M.Eng. degrees in mechanical engineering from the School of Mechatronics Engineering, Zhejiang University, Hangzhou, China. He is currently working toward the Ph.D. degree in biomedical engineering with the Department of Biomedical Engineering, Faculty of Engineering Technology, University of Twente, The Netherlands.

His research interest includes dynamic on-body and in-body compression systems.



Guangda Qiao received the B.E. and M.E. degrees in mechanical engineering from the School of Mechatronics Engineering, Changchun University of Technology, Changchun, China, in 2019 and 2022, respectively. He is currently working toward the Ph.D. degree in mechatronics engineering with the School of Mechanical Engineering, Zhejiang University, Zhejiang, China.

His research interests include piezoelectric actuators, medical devices, and triboelectric

nanogenerators.



Guofang Gong received the Ph.D. degree in mechanical engineering from the China University of Mining and Technology, Beijing, China, in 2000.

He is a Doctoral Tutor with the Institute of Mechatronics and Control Engineering, Zhejiang University, Hangzhou, China. He has implemented one project from the National Natural Science Foundation and three projects from 863 Hi-tech Project. He has authored or coauthored more than 100 journal and conference papers.

His research interests include intelligent tunnel boring machines, electrohydraulic control systems, tribological characteristics of materials and their applications in hydraulic components.

Dr. Gong is the recipient of the National Scientific and Technological Progress Prize (first class) and the National Scientific and Technological Progress Prize (second class).



Huayong Yang received the Ph.D. degree in fluid power transmission and control from the University of Bath, Bath, U.K., in 1988.

He is currently the Cheung Kong Scholar Chair Professor with Zhejiang University, Hangzhou, China. His research interests include intelligent tunnel boring machines, motion control and energy saving of mechatronic systems, microfluidic devices and systems, and R&D of fluid power components.

Prof. Yang is the recipient of the National Scientific and Technological Progress Prize (first class) and the National Scientific and Technological Progress Prize (second class). He has also received the National Outstanding Researcher of the Natural Science Foundation of China Award. He is the Editor-in-Chief of *Journals of Zhejiang University-SCIENCE A* (Applied Physics and Engineering) and *Bio-Design and Manufacturing*. He is a member of the Chinese Academy of Engineering.



Chun-Yi Su (Senior Member, IEEE) received the Ph.D. degree in control engineering from the South China University of Technology, Guangzhou, China, in 1990.

After a seven-year stint with the University of Victoria, Victoria, BC, Canada, he joined Concordia University, Montreal, QC, Canada, in 1998. He is currently a Professor and Honorary Concordia University Research Chair with Gina Cody School of Engineering and Computer Science, Concordia University. His research inter-

est includes control theory and its applications to various mechanical systems.

Prof. Su has served as an Associate Editor for the IEEE TRANSACTIONS ON AUTOMATIC CONTROL and IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY. He has been on the editorial board of several journals, including *Mechatronics* (IFAC). He is a Distinguished Lecturer of the IEEE Robotics and Automation Society and has served for many conferences as an organizing committee member in various capacities, including the general chair and program chair.



Fuzhou Niu (Member, IEEE) received the B.S. degree in mechanical design and manufacturing engineering from the Department of Precision Machinery and Instrumentation, University of Science and Technology of China, Hefei, Anhui, China, in 2010, and the Ph.D. degrees in robotics and biomedical engineering from the University of Science and Technology of China and City University of Hong Kong, Hong Kong, China, in 2017 and 2018, respectively.

He joined Suzhou University of Science and Technology in 2018. He is currently working as an Associate Professor with the School of Mechanical Engineering, Suzhou University of Science and Technology, Suzhou, China. His research interests include micro/nano-manipulation, microrobot systems, and bio/micro-robotic applications.



Yueyue Liu (Member, IEEE) received the M.S. and Ph.D. degrees in control science and engineering from the South China University of Technology, Guangzhou, China, in 2017 and 2021, respectively.

He is currently working as an Associate Professor with the School of Internet of Things Engineering, Jiangnan University, Wuxi, China. His research interests include microrobots, autonomous navigation, motion planning, and non-linear control.



Dong Han received the Dr. Eng. degree in mechanical engineering from Tokyo Institute of Technology (Tokyo Tech), Tokyo, Japan, in 2018.

From 2018 to 2020, he joined the Laboratory for Future Interdisciplinary Research of Science and Technology (FIRST), Tokyo Tech, as an Assistant Professor. In 2020, he worked as a Research Professor with the School of Mechanical Engineering and State Key Laboratory of Fluid Power and Mechatronic System, Zhejiang University, Zhejiang, China. His research interests include intelligent tunnel boring machines, electrohydraulic control systems, electromagnetic, piezoelectric and functional fluid drive systems, and medical device development.